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Comparison of OWAS, RULA and REBA for assessing potential work-related musculoskeletal disorders

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ABSTRACT

This study compared three representative observational methods for assessing musculoskeletal loadings: Ovako Working Posture Analysis System (OWAS), Rapid Upper Limb Assessment (RULA), and Rapid Entire Body Assessment (REBA). The comparison was based on 209 cases of upper-body musculoskeletal disorders (MSDs) diagnosed by medical doctors. The most awkward/stressful posture in each participant's tasks was assessed using these techniques. Postural loadings were rated more highly by the RULA than by the OWAS and REBA ($p < 0.01$). The chi-square test and logistic regression analysis showed that only RULA grand score and action level, and REBA action level were associated with MSD work-relatedness ($p < 0.01$, $p < 0.05$, and $p < 0.05$, respectively). The percentage concordant values of the logistic model for the RULA grand score and action level were 52.4% and 44.8%, respectively, while the percentage concordant value for the REBA action level was 22.1%. Therefore, the RULA may be the best system for estimating the postural loads and work-relatedness of MSDs.

Relevance to industry: Work-related musculoskeletal disorders are the leading cause of workplace disability in the developed countries. For preventing the disorders, quantification of musculoskeletal loads is required.

1. Introduction

In 2019, 9440 cases of work-related musculoskeletal disorders (WMSDs) were reported in Korea, representing an increase of 2725 cases (40.6%) from the 6715 cases reported in the previous year. The cases accounted for approximately two-thirds (67.3%) of all occupational diseases in that year (Ministry of Employment and Labor, 2020). In the USA, WMSDs amounted to 29–35% of occupational injuries and illness involving days away from work in 1992–2010. The total cost, including direct and indirect, was estimated to be $2.6 billion in 2007 (Bhattacharya, 2014).

The implementation of intervention programs for reducing exposure to musculoskeletal disorder (MSD)-related risk factors is the best-known prevention strategy (Burdorf, 2010; Chiasson et al., 2012; Silverstein and Clark, 2004). To implement certain intervention programs, musculoskeletal loadings should be evaluated. Among the various assessment methods, observational techniques have been used more frequently. They are inexpensive, easy to use, flexible, and do not interfere with workers' tasks or the jobs being performed (Chiasson et al., 2012; David, 2014; Genaidy et al., 1994; Gómez-Galán et al., 2017; Sukadarin et al., 2016; Takala et al., 2010).

Many studies have compared observational methods for assessing MSD-related risk factors. The Ovako Working Posture Analysis System (OWAS) (Karhu et al., 1977), Rapid Upper Limb Assessment (RULA) (McAtamney and Corlett, 1993), and Rapid Entire Body Assessment (REBA) (Hignett and McAtamney, 2000) are the three techniques that have been frequently compared in the literature and applied in industries because they make it possible to identify physical effort by posture, force, and static or repetitive load (Ansari and Sheikn, 2014; Choobineh et al., 2004; Colim et al., 2020a,b; Dianat et al., 2020; Gómez-Galán et al., 2017; Gómez-Galán et al., 2020; González et al., 2003; Hita-Gutiérrez et al., 2020; Joshi and Deshpande, 2019; Kee, 2020; Kee et al., 2020; Lowe et al., 2019; Lu et al., 2016; Madani and Dababneh, 2016; Massaccesi et al., 2003; Roma-Liu, 2014; Sane et al., 201). However, most previous studies have simply summarized and compared the existing observation techniques, based on their general characteristics, including correspondence with a valid reference, scales for posture classification, main functions, association with MSDs,

Abbreviations: COMWEL, Korea Workers' Compensation & Welfare Service; LUBA, Loading on the Upper Body Assessment; MSDs, musculoskeletal disorders; NERPA, Novel Ergonomic Posture Assessment; OWAS, Ovako Working Posture Analysis System; REBA, Rapid Entire Body Assessment; RULA, Rapid Upper Limb Assessment; WMSDs, work-related musculoskeletal disorders.

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inter-observer repeatability, potential users, and exposure factors assessed such as postures or body regions, load/force, and movement frequency (David, 2014; Joshi and Deshpande, 2019; Roman-Liu, 2014; Sukadarin et al., 2016; Takala et al., 2010). Some studies compared the three methods based on their assessment results (Bartnicka, 2015; Chiasson et al., 2012; Enez and Nalbantoğlu, 2019; Kee, 2020; Kee et al., 2020; Kee and Karwowski, 2007; Kong et al., 2017; Yazdanirad, 2018). For example, Kee (2020) and Kee et al. (2020) empirically compared the three observational methods based on the perceived discomfort and maximum holding times measured from experiments, respectively.

Rathore et al. (2020) reported that the working posture variables collected using the REBA and the musculoskeletal symptoms for the different body regions were significantly correlated through the logistic regression models. Shuval and Donchin (2005) and Yazdanirad et al. (2018) investigated the relationship between the prevalence of the upper extremity symptoms as evaluated using questionnaires and assessment results or scores, with the former study based on the RULA and the latter on the RULA, Loading on the Upper Body Assessment (LUBA) (Kee and Karwowkski, 2001), and Novel Ergonomic Posture Assessment (NERPA) (Sanchez-Lite et al., 2013). Although the validity of an ergonomic risk assessment technique can be appraised by evaluating how well its risk estimates are associated with the symptoms of MSDs (Takala et al., 2010), none of the previous studies were based on objective MSDs diagnosed by medical doctors. Furthermore, although objective and subjective MSDs do not necessarily correlate, the studies by Rathore et al. (2020), Shuval and Donchin (2005), and Yazdanirad et al. (2018) used only subjective MSD symptom data obtained using the Cornell Musculoskeletal Discomfort Questionnaire (Cornell University Ergonomics Web, 2003) and the Nordic Musculoskeletal Questionnaire (Kuorinka et al., 1987), respectively.

Therefore, this study compared the three commonly used observational techniques (OWAS, RULA, and REBA) by assessing the association between the work-related musculoskeletal loadings determined using these techniques and objective MSD data. The author analyzed the MSDs of patients who were diagnosed by medical doctors and who applied for medical care due to industrial accidents to the Korea Workers' Compensation & Welfare Service (COMWEL), which is a quasi-governmental agency that functions under the Ministry of Employment and Labor of Korea and implements various social security and labor welfare services and programs.

1.1. OWAS, RULA, and REBA

The OWAS was developed by a Finnish steel company of Ovako Oy (Karhu et al., 1977). The method was based on ratings of working postures taken in several divisions of the steel factory performed by 32 experienced steel workers and international ergonomists. The OWAS identifies four work postures for the back, three for the arms, seven for the lower limbs, and three categories for the weight of load handled or amount of force used. The technique classifies combinations of these four categories by the degree of their impact on the musculoskeletal system for all posture combinations. The degrees of the assessed harmfulness of these posture–load combinations are grouped into four action categories, which indicate the urgency for the required workplace interventions: action category 1, normal postures which do not need any special attention, except in some special cases; action category 2, postures must be considered during the next regular check of working methods; action category 3, postures need consideration in the near future; and action category 4: postures need immediate consideration.

The RULA was proposed to provide a quick assessment of the loading on the musculoskeletal system due to the postures of the neck, trunk and upper limbs, muscle function, and external loads exerted. Based on the grand score of its coding system, four action levels, which indicate the level of intervention required to reduce the risks of injury due to physical loading on the worker, were suggested (McAtamney and Corlett, 1993): action level 1, posture is acceptable if it is not maintained or repeated for long periods; action level 2, further investigation is needed and changes may be needed; action level 3, investigation and changes are required soon; and action level 4, investigation and changes are required immediately.

The REBA is a postural analysis tool designed to be sensitive to the type of unpredictable working postures found in health care and other service industries. The posture classification system, which included the upper arms, lower arms, wrist, trunk, neck, and legs, was based on the body part diagrams from the RULA. The method reflected the extent of the external load/forces exerted, muscle activity caused by static, dynamic, rapid changing or unstable postures, and coupling effect. Unlike the OWAS and RULA, this technique provided five action levels for evaluating the level of corrective actions (Hignett and McAtamney, 2000): action level 0, corrective action including further assessment is not necessary; action level 1, corrective action including further assessment may be necessary; action level 2, corrective action including further assessment is necessary; action level 3, corrective action including further assessment is necessary soon; and action level 4, corrective action including further assessment is necessary now.

2. Material and methods

2.1. MSD data collection

Data on 229 cases of MSDs for the 5-year period from 2015 to 2019 were collected; these patients were medically diagnosed by doctors, and the individuals had applied to the COMWEL for medical care attributed to industrial accidents. Data were obtained through the author's personal communications with the health and/or safety managers of 18 companies in Korea, including automotive and automotive parts manufacturing, and construction industries (Table 1). These industries are associated with the most frequent WMSDs in Korea (Ministry of Employment and Labor, 2020). It is difficult to classify tasks performed in the construction industry by each worker because it is common for a worker to perform several different tasks in construction sites. The tasks for the 40 cases in the construction industry of Table 1 include wood working, rebar work, pipe installation, electric work, scaffolding installation, welding, interior work, and others. The tasks in automotive and automotive parts manufacturing consist of 100 manufacturing/assembly, 31 maintenance and repair, 13 transportation, 13 metal processing, 6 welding, and 6 other tasks (3 clerical work, 2 inspection and packaging, and 1 food service).

Videos or photographs of the applicants performing tasks on site during their regular working hours were also received, together with the MSD data. Before filming the working scenes, the participants were informed of its purpose, following which they provided written informed consent. Of the 229 cases of MSDs, 20 cases involving MSDs of the lower limbs were excluded from analysis, because the RULA does not have enough postural classifications to assess postural loads of the lower limb appropriately. While the OWAS and REBA have seven and six postural codes for the lower limb, respectively, the RULA has only two postural categories according to whether the legs and feet are well supported and in an evenly balanced posture (Hignett and McAtamney, 2000; Karhu et al., 1977; McAtamney and Corlett, 1993).

Overall, 148 applications were approved as WMSDs by the COMWEL (automotive and automotive parts, 115; and construction, 33), whereas for 61 cases, the COMWEL determined that the MSDs were attributed to factors other than work-related factors (automotive and automotive parts, 54; and construction, 7). Approved applicants would receive a wage for the leave of absence and cost of medical treatment during the medical care allowed by the COMWEL. Details of the applicants' sex, age, and industrial sector according to whether applications were approved or rejected by the COMWEL are summarized in Table 1.

Table 1
Applicants' sex, age, and industrial sector.

WMSDs	Sex		Average age (years)		Industrial sector	
	Female	Male	Female	Male	Automotive and automotive parts	Construction
Approved	17	131	52.2 (8.9)	50.9 (11.1)	115	33
Rejected	13	48	50.5 (6.1)	49.0 (11.2)	54	7
Total	30	179	51.4 (7.7)	50.4 (11.1)	169	40

Data are presented as number or mean (standard deviation).
WMSDs, work-related musculoskeletal disorders.

2.2. Data analysis

First, the author selected a posture for each case that reflected the most awkward or stressful task among those performed by the applicant based on the videos and photographs. The posture was chosen from tasks with a working duration of 1 or more hours in the daily routine. Second, the author assessed postural loads for the 209 selected postures using the OWAS, RULA, and REBA posture classification schemes. This resulted in five postural load scores for each posture: an action category for the OWAS, a grand score and an action level for the RULA, and a score and an action level for the REBA. It was assumed during the analysis that the OWAS action category and RULA and REBA action levels were on the ordinal scale, and the RULA grand and REBA scores represented an interval scale. Third, the three assessment techniques were compared by plotting the postural loads of each technique, and by using the Wilcoxon signed-rank test, quadratic weighted κ analysis, and Spearman's and Pearson's correlation analyses. The Wilcoxon signed-rank test was used to evaluate whether the mean ranks for the techniques' risk levels differ. The quadratic weighted κ analysis was used to examine agreements between the techniques' risk levels. The Spearman's and Pearson's correlation analyses were employed to investigate the relationships between the techniques' risk levels and scores including RULA grand score and REBA score, respectively.

Each technique has its own scales of action levels or risk categories that guide the risk levels and the indication of urgency of more detailed assessments and improvements (Hignett and McAtamney, 2000; McAtamney and Corlett, 1993). The OWAS and RULA classify postural loads related to the urgency of corrective actions into four action categories and action levels, respectively; these terms have similar meanings. The REBA classifies postural loads into five action levels, which have meanings that are slightly different from those of the action categories/levels of the OWAS and RULA. To enable an effective comparison, the five REBA action levels were regrouped into four categories, with consideration of the meanings of the action categories/levels for the three techniques. The four new REBA action levels were as follows: action levels 1 (originally action level 0), 2 (originally action levels 1 and 2), 3 (originally action level 3), and 4 (originally action level 4). Finally, the chi-square tests and logistic regression analysis were selected to investigate the significances of musculoskeletal loadings assessed using the three techniques on MSDs and the relationships between the MSDs and the musculoskeletal loadings, respectively. All statistical analyses were conducted using SAS (SAS Inc., Cary, NC, USA) and Microsoft Excel (Microsoft Co., Redmond, WA, USA).

3. Results

3.1. Body parts for MSDs

The distribution of the body parts affected by MSDs is shown in Table 2. In 10 cases, two body parts were involved. Shoulder disorders were the most frequent, independent of the industrial sectors, followed by problems in the low back. There were no significant differences in the number of MSDs between the hand/wrist, elbow, and neck disorders.

Table 2
Body parts involved in MSDs.

	Wrist/hand	Elbow	Shoulder	Low back[a]	Neck	Total
Automotive and automotive parts	7 (4.0[b])	14 (8.0)	86 (48.8)	59 (33.5)	10 (5.7)	176 (100.0)
Construction	3 (7.0)	1 (2.3)	25 (58.1)	12 (27.9)	2 (4.7)	43 (100.0)
	10 (4.6)	15 (6.8)	111 (50.7)	71 (32.4)	12 (5.5)	219 (100.0)

MSDs, musculoskeletal disorders.
[a] One case of thoracic vertebrae is included in the low back.
[b] The numbers in parentheses represent percentage values relative to the horizontal sum.

3.2. Comparisons by risk levels

The action categories of the OWAS, as related to the RULA action levels, are presented in Fig. 1a. The action categories and levels for only 37 postures were the same (18 action category or level 2 classes, 18 action category or level 3 classes, and one action category or level 4 class), while those for the remaining 172 postures were different. The percentage of agreement between the two methods was 17.7%. The OWAS underestimated the postural loads for 172 postures, as compared to the RULA. There were 120 postures with a difference of one risk level, 37 with a difference of two levels, and 15 with a difference of three levels (Table 3). The Wilcoxon signed-rank test showed that the risk levels as determined using the OWAS were statistically significantly lower than those determined using the RULA ($p < 0.01$). The Spearman correlation coefficient between the OWAS action categories and RULA action levels was not high ($r = 0.56$, $p < 0.01$).

The action categories and levels as determined using the OWAS and REBA were the same for 75 postures, indicating a coincidence rate of 35.9%. Compared to the REBA, the OWAS underestimated the postural loads for 105 postures but overestimated the loads for 29 postures (Fig. 1b). The Wilcoxon signed-rank test indicated that the OWAS generally underestimated postural stress compared to the REBA ($p < 0.01$). The Spearman correlation coefficient between the OWAS action categories and REBA action levels was also not high ($r = 0.43$, $p < 0.01$).

The RULA estimated postural loads for 121 (57.9%) postures more highly than the REBA, which was statistically significant according to the Wilcoxon signed-rank test ($p < 0.01$) (Fig. 1c). While the RULA underestimated just two postures at a one-level lower action level, it evaluated 83 postures at a one-level higher action level and 38 postures at a two-level higher action level than the REBA (Table 3). Eighty-six postures were estimated to have the same action levels as the REBA, which corresponded to an inter-technique reliability of 41.1%. The Spearman correlation coefficient between the RULA and REBA action levels was not high ($r = 0.45$, $p < 0.01$). The Pearson correlation coefficient between the RULA grand score and REBA score was also not high ($r = 0.59$, $p < 0.01$).

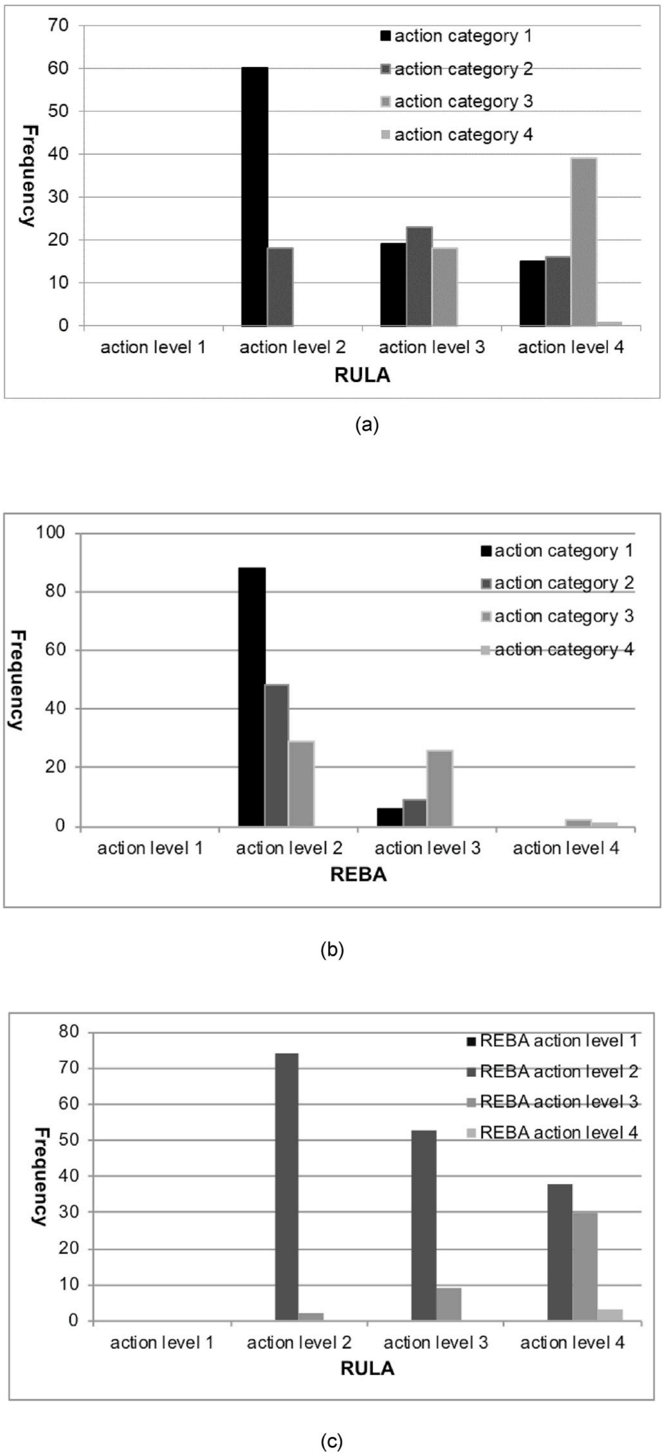

Fig. 1. Distribution of risk categories by techniques. (a) Ovako Working Posture Analysis System (OWAS) and Rapid Upper Limb Assessment (RULA). (b) OWAS and Rapid Entire Body Assessment (REBA). (c) RULA and REBA.

Table 3
Differences in risk levels between pairs of assessment methods.

	0 level	One level	Two levels	Three levels
OWAS and RULA	37	120	37	15
OWAS and REBA	75	128	6	0
RULA and REBA	86	85	38	0

OWAS, Ovako Working Posture Analysis System; RULA, Rapid Upper Limb Assessment; REBA, Rapid Entire Body Assessment.

3.3. Risk categories by the techniques

The risk categories by the OWAS, RULA, and REBA are illustrated in Fig. 2. The OWAS and REBA assessed 151 (72.2%) and 165 (78.9%) postures as action categories or levels 1 or 2, respectively, which indicated acceptable postures or no requirement for corrective changes. The RULA evaluated 133 (63.6%) postures as action levels 3 (62 postures) or 4 (71 postures), which indicated the need for rapid or immediate corrective actions to reduce postural loads.

3.4. Quadratic weighted κ analyses

A quadratic weighted κ analysis was adopted to investigate the inter-technique agreement between the risk levels by the three methods. The inter-technique agreements between the OWAS and RULA, the OWAS and REBA, and the RULA and REBA were fair ($\kappa = 0.29$, $\kappa = 0.32$, and $\kappa = 0.23$, respectively).

3.5. Significance of risk levels on MSDs

Distribution of the risk levels as determined using the different assessment techniques according to whether the applied MSDs were approved as WMSDs is illustrated in Fig. 3. The OWAS did not show significant differences in the risk levels between the MSDs approved (WMSDs) and rejected (MSDs not approved as WMSDs) by the COMWEL, while the RULA identified significant differences in the risk levels between these categories. In the RULA, the percentage of postures with high postural loading assessed as action level 4 was markedly higher for WMSDs than for rejected MSDs, but the values for the low postural loadings assessed as action levels 2 and 3 were inversely higher for rejected MSDs than for WMSDs. Although there were differences in the REBA action levels between the WMSDs and rejected MSDs, as stated above, the REBA underestimated three-fourths of the 209 postures as action level 2, irrespective of the work-relatedness of MSDs.

The significance of the risk levels, determined using the three assessment techniques, according to whether the applied MSDs were considered WMSDs, was tested using the chi-square tests. To enhance the reliability of these tests, some risk categories were amalgamated to have at least five elements in each cell of the contingency table. Thus, the one OWAS action category 4 class was merged into action category 3, and the three REBA action level 4 classes were combined into action level 3. While the OWAS action category was not associated with the work-relatedness of MSDs ($p > 0.10$), the RULA and REBA action levels were significantly related to the work-relatedness of MSDs ($p < 0.05$). In addition to the action category or level, the RULA grand score, with a range of 3–7, and the REBA score, with a range of 2–13, were also

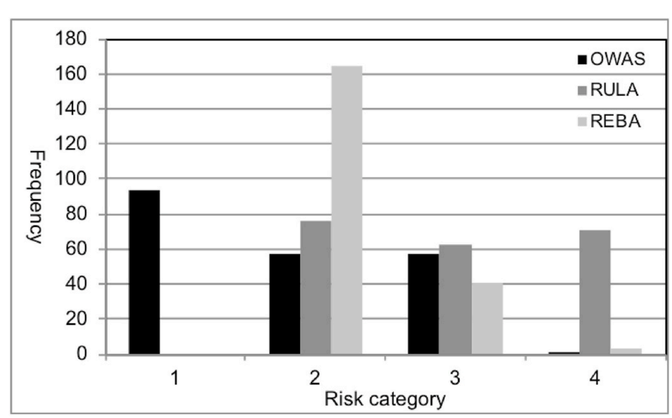

Fig. 2. Distribution of risk categories by techniques. OWAS, Ovako Working Posture Analysis; System; RULA, Rapid Upper Limb Assessment; REBA, Rapid Entire Body Assessment.

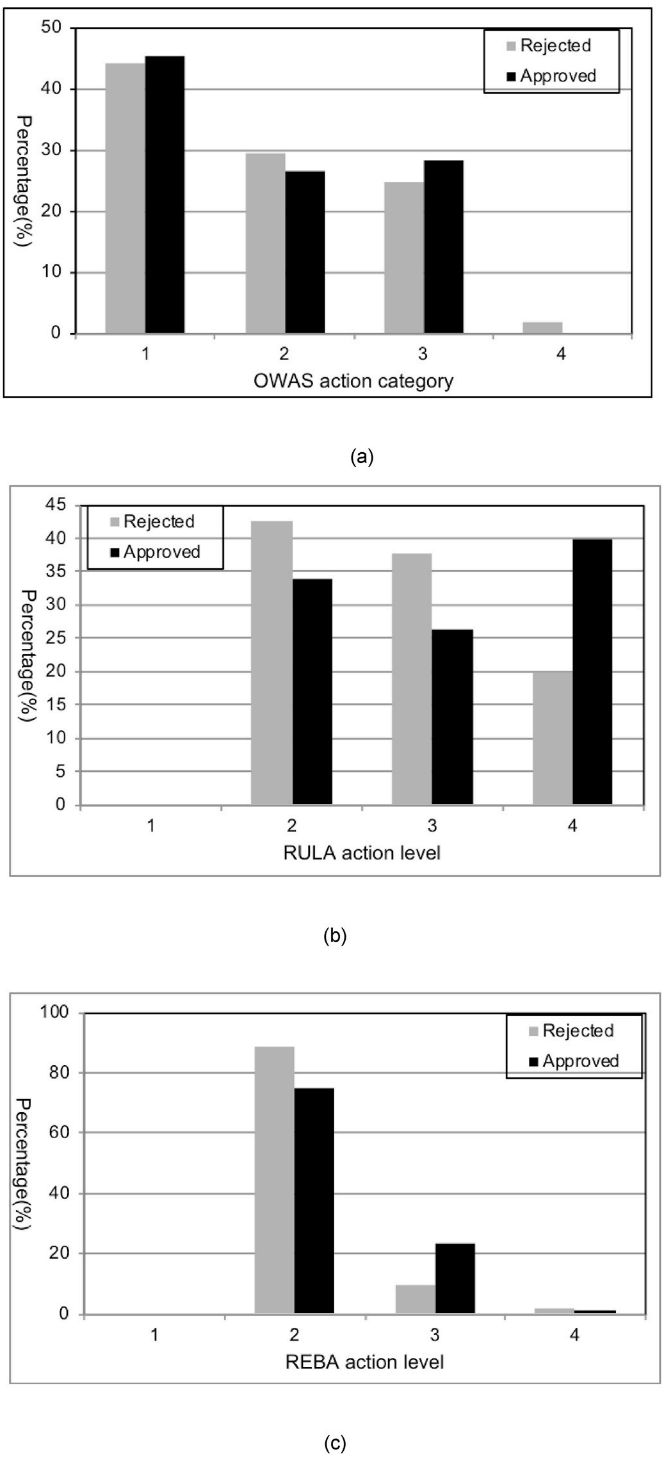

Fig. 3. Distribution of risk levels by methods and work-relatedness of MSDs. (a) Ovako Working Posture Analysis System (OWAS). (b) Rapid Upper Limb Assessment (RULA). (c) Rapid Entire Body Assessment. Values in the vertical axis represent the percentage of each risk level to the sum of the cases for the work-related musculoskeletal disorders and the rejected musculoskeletal disorders.

subjected to the chi-square analyses. The RULA grand score was a significant determinant of whether the MSDs were considered to be work-related ($p < 0.01$), but the REBA score was not ($p > 0.10$).

3.6. Logistic regression analysis

Logistic regression analyses were performed considering the RULA grand score and action level and the REBA action level, which were significant in the chi-square analyses, as independent variables. The dependent variable was whether the MSDs were approved to be work-related (i.e., WMSDs vs rejected MSDs). Three separate logistic regression analyses were conducted for each independent variable (Table 4). To increase the reliability of the analyses, the three REBA action level 4 classes were recategorized as action level 3. Here, the RULA grand score was assumed to be the interval scale, with characteristics of a continuous number.

The analyses revealed that the RULA grand score and action level and the REBA action level were statistically significant determinants of whether an MSD would be considered a WMSD ($p < 0.01$, $p < 0.05$, and $p < 0.05$, respectively). An increase of 1 point in the RULA grand score increased the odds ratio for the assessment of MSDs as WMSDs by approximately 1.36 times. The odds ratios for the RULA and REBA action levels were calculated with reference to level 2, which was the minimum action level in the RULA and REBA analyses. RULA action level 4 and REBA action level 3 almost tripled the probability that the MSDs would be approved as work-related (2.56 and 2.57 times, respectively), compared to this probability for the RULA and REBA action level 2. The percentage concordant values of the logistic regression models for the RULA grand score and action levels were slightly higher (52.4% and 44.8%, respectively), while the value for the REBA action level was relatively low (22.1%).

4. Discussion

This study compared three observational techniques for assessing whether the MSDs are work-related using objective MSD data. Our findings indicated that the RULA might be the best system for estimating the postural loads and work-relatedness of MSDs because (1) the RULA generally rated the 209 postures assessed as being more stressful than did the OWAS and REBA, and (2) the RULA grand score and action level were more significantly associated with the decision regarding the work-relatedness of MSDs than were the OWAS and REBA. These are in line with the results of Kee (2020), Kee et al. (2020), Kong et al. (2017), and Yazdanirad et al. (2018).

Most previous comparative studies compared or evaluated ergonomic assessment methods based on the generic characteristics, results of evaluation of musculoskeletal loadings or risk levels, body discomfort, maximum holding time, subjective musculoskeletal symptoms, and others (Bartnicka, 2015; Chiasson et al., 2012; David, 2005; Enez and Nalbantoğlu, 2019; Joshi and Deshpande, 2019; Kee, 2020; Kee et al., 2020; Kee and Karwowski, 2007; Rathore et al., 2020; Roman-Liu, 2014; Roman-Liu et al., 2013; Shuval and Donchin, 2005; Takala et al., 2010; Yazdanirad et al., 2018). Comparisons by Yazdanirad et al. (2018), and evaluations by Shuval and Donchin (2005) and Rathore et al. (2020)

Table 4
WMSDs by musculoskeletal loadings.

Independent variable	N	OR	95% CI	% Concordance
RULA grand score				52.4
Continuous (per 1 point)	209	1.36	1.10-1.68	
RULA action level				44.8
2	76	1		
3	62	0.88	0.44–1.78	
4	71	2.56	1.17–5.58	
REBA action level				22.1
2	164	1		
3	44	2.57	1.08–6.14	

*OR, odds ratio; CI, confidence interval; RULA. Rapid Upper Limb Assessment; REBA, Rapid Entire Body Assessment; WMSDs, work-related musculoskeletal disorders.

were based on musculoskeletal symptoms, but the symptoms were subjectively reported via questionnaires. Compared to these previous studies, this study had the advantage of being based on real MSD data diagnosed by doctors. In Korea, workers with MSDs, with a doctor's certificate, can apply to the COMWEL for medical treatment for these disorders. The COMWEL determines whether the applicants' MSDs are work-related by deliberation. If the COMWEL judges the applicants' MSDs to be work-related, it pays the person's wage for the leave of absence and the cost of medical treatment during the medical care of the applicants.

The results showed that while the OWAS and REBA indicated that 72.2% and 78.9% of the postures assessed in this study had low risk levels, with action categories or levels of 1 or 2, respectively, the RULA pointed out that 63.6% of the postures had high postural loadings, with action levels of 3 or 4. The RULA did not identify any postures as action level 1, while the OWAS and REBA considered one and three postures as action category or level 4, respectively. This was in agreement with several previous studies, which suggested that of the three observation techniques, the RULA overestimated postural loads compared to the OWAS and REBA (Chiasson et al., 2012; Kee, 2020; Kee et al., 2020; Kee and Karwowski, 2007; Sahu et al., 2013; Saliha et al., 2017; Upasana and Vinay, 2017). The rates of concordance between the OWAS and RULA, the OWAS and REBA, and the RULA and REBA were 17.7%, 35.9%, and 41.1%, respectively. These results were slightly similar to the concordance rates of 29.2%, 48.2%, and 54.8%, respectively, reported by Kee and Karwowski (2007), and 16.7%, 8.3%, and 33.3%, respectively, reported Kee (2020). Enez and Nalbantoğlu (2019) indicated that the agreement rate between the OWAS and REBA was 29.1% in forestry timber harvesting. Chiasson et al. (2012) reported that the inter-technique reliability between the RULA and REBA was 73.7%, which was markedly different from the finding of the present study.

The quadratic weighted κ between the risk levels by the three techniques ranged from 0.23 to 0.32, which corresponded to "fair," and the concordance rates were not high (17.7%–41.1%). In addition, the Spearman correlation coefficients between them were also not high (0.43–0.56). These indicated that the risk levels by the three techniques were not strongly correlated.

The OWAS and REBA have some characteristics in common. First, they assessed about three-fourths of the 209 postures as being in action categories or levels of 1 and 2. Second, they could not properly distinguish the work-relatedness of MSDs on the basis of only risk levels (Fig. 3). The chi-square tests and logistic regression analysis revealed that the OWAS could not accurately predict whether the applied MSDs were work-related. Although the REBA action level appeared to be statistically significantly associated with the work-relatedness of MSDs, through the chi-square test and logistic regression analysis, it could also not appropriately evaluate the work-relatedness of MSDs based on the REBA action level alone. This was because the REBA assessed most postures (78.9%) with the low risk level, not requiring rapid or immediate corrective actions, irrespective of the work-relatedness of MSDs, and the REBA identified only three cases of action level 4 (Figs. 1–3).

The chi-square test and logistic regression analysis demonstrated that the RULA grand score and action level and the REBA action level were significantly associated with whether the applied cases of MSD were attributed to work-related factors. However, the concordance rates for the logistic regression analysis were low to medium, rather than high (22.1–52.4%). This may be due to several reasons. First, the RULA and REBA do not properly assess the work-related musculoskeletal load, which is known to be the main risk factor in the development of MSDs (Coury and Padula 2002; Kumar, 2001; Sande et al., 2001). The REBA, with a low concordance (22.1%), as well as the RULA, with a slightly higher concordance (44.8% for the action level and 52.4% for the RULA grand score), may have problems related to the scoring system and categorization of the action levels. Second, it is difficult to estimate the work-relatedness for different MSDs affecting various body parts and being developed due to various factors using a simple ergonomic method only. This implies that several methods are needed for the proper evaluation of musculoskeletal loading and for predicting disorders, rather than using a single best technique. Thus, new techniques should be developed for specific body parts and/or MSDs, rather than for the whole body and all MSDs, as most existing methods do. Third, MSDs are characterized as multi-factorial, i.e., caused by the physical as well as psychosocial/organizational and individual factors (Armstrong et al., 1993; Bongers et al., 1993; Devereux et al., 1999; 2002; van der Beek et al., 1998; Winkel and Mathiassen, 1994). Ergonomics methods, including the OWAS, RULA and REBA, have been developed mainly for assessing the physical or work-related factors, and not for evaluating the psychosocial/organizational and individual factors.

Although the RULA focused on the classification of the upper limb postures rather than the lower limb postures, which has just two postural codes for the leg postures, and the OWAS and REBA were developed for assessing the whole-body postures, this study comparing the three methods can be justified as follows. First, of the various observational techniques, the three methods have been cited in the literature and applied in industrial sites most frequently (Gómez-Galán et al., 2017, 2020; Hita-Gutiérrez et al., 2020; Madani and Dababneh, 2016). Many studies adopted the RULA for assessing postural loads in the agriculture, forestry and fishing, mining and quarry, construction, and shipbuilding industries as well as general hospitals requiring frequent unstable lower limb postures such as squatting and kneeling (Gómez-Galán et al., 2020; KOSHA, 2005). Second, previous studies demonstrated that the RULA evaluated postural loads even for postures with unstable leg movements more accurately or highly than the OWAS and REBA. Kong et al. (2017) reported that the RULA had higher agreements with ergonomic expert assessments for various agricultural tasks than the OWAS and REBA. Kee and Karwowski (2007) pointed out that regardless of industry, tasks type, and leg postural balance, the RULA more highly estimated posture-related risks for 301 different postures drawn from the iron and steel, electronics, automotive, and chemical industries as well as general hospitals than the OWAS and REBA. Kee (2020) and Kee et al. (2020) also reported high estimates from the use of RULA for postural loads of postures with unstable leg postures.

This study had some limitations. It was based on an analysis of 209 cases of MSDs collected from two industries in the manufacturing and construction sectors. Further studies using more MSD data from varying industries are required to obtain more generalizable results.

5. Conclusions

Following the suggestion that the validity of an observational technique can be assessed by its results in association with MSDs (Takala et al., 2010), this study compared the OWAS, RULA and REBA based on real MSD data obtained from industrial sites. Of the three techniques compared, the RULA may be better for assessing the postural loads as well as the work-relatedness of MSDs. This was on the basis of the findings of this study indicating the RULA's overestimation for postural loads, and high association with the work-relatedness of MSDs. The result is expected to be helpful in selecting an appropriate observational technique when evaluating postural loads.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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